

Grain 1: Order-Level Analysis (30_Orders Enriched)

Predictor Screening					
Predictor	order_total_kes			Rank ^	Copy Selected
	Contribution	Portion			
size_tier	1.332e+10	0.9246		1	
shop_type	445985793	0.0310		2	
credit_terms	240504687	0.0167		3	
distance_km	142496289	0.0099		4	
road_quality_index	95046029	0.0066		5	
avg_stops_planned	92524948	0.0064		6	
prior_fmcs_experience_years	69437091	0.0048		7	

Grain 2: Line-Item SKU Analysis (16_daily_sales_Enriched)

Predictor Screening					
Predictor	line_amount_kes			Rank ^	Copy Selected
	Contribution	Portion			
unit_price_kes	821712558	0.5084		1	
unit_cost_kes	739408587	0.4575		2	
pack_size_ml	41134350	0.0255		3	
shelf_life_days	12313566	0.0076		4	
category	1638835.2	0.0010		5	
cold_chain_required	0	0.0000		6	

Grain 3: Visit Conversion Analysis (17_Daily_Visits_Enriched_VisitLevel)

Predictor Screening					
Predictor	outcome			Rank ^	Copy Selected
	Contribution	Portion			
visit_duration_min	159.058	0.3583		1	
shop_type	74.891	0.1687		2	
distance_km	54.344	0.1224		3	
credit_terms	52.852	0.1190		4	
size_tier	47.708	0.1075		5	
prior_fmcs_experience_years	28.768	0.0648		6	
road_quality_index	26.338	0.0593		7	

Grain 4: Delivery Delay Performance (18_Delivery_Log_Enriched_DeliveryLevel)

Predictor Screening [Copy Selected](#)

Predictor	Contribution	Portion	delay_days	Rank ^
order_total_kes	64.0396	0.6645		1
route_id	11.4468	0.1188		2
item_count	8.6451	0.0897		3
territory_id	4.7832	0.0496		4
delivered_pct_of_order	4.3478	0.0451		5
capacity_kg	2.6950	0.0280		6
cold_chain_temp_excursion	0.4219	0.0044		7
fuel_type	0.0000	0.0000		8

Grain 5: Stockout Duration Analysis (29_Stockouts_Enriched_StockoutLevel)

Response duration_days

Singularity Details

Term	Details
Intercept	=0.28571*category[Drinking] - 0.28571*category[Greek] + 0.07143*shelf...

Whole Model

Effect Summary

Source	Logworth	PValue
category	4.572	0.00003
cause	0.850	0.14121
cold_chain_required	.	.
shelf_life_days	.	.
scope	.	.

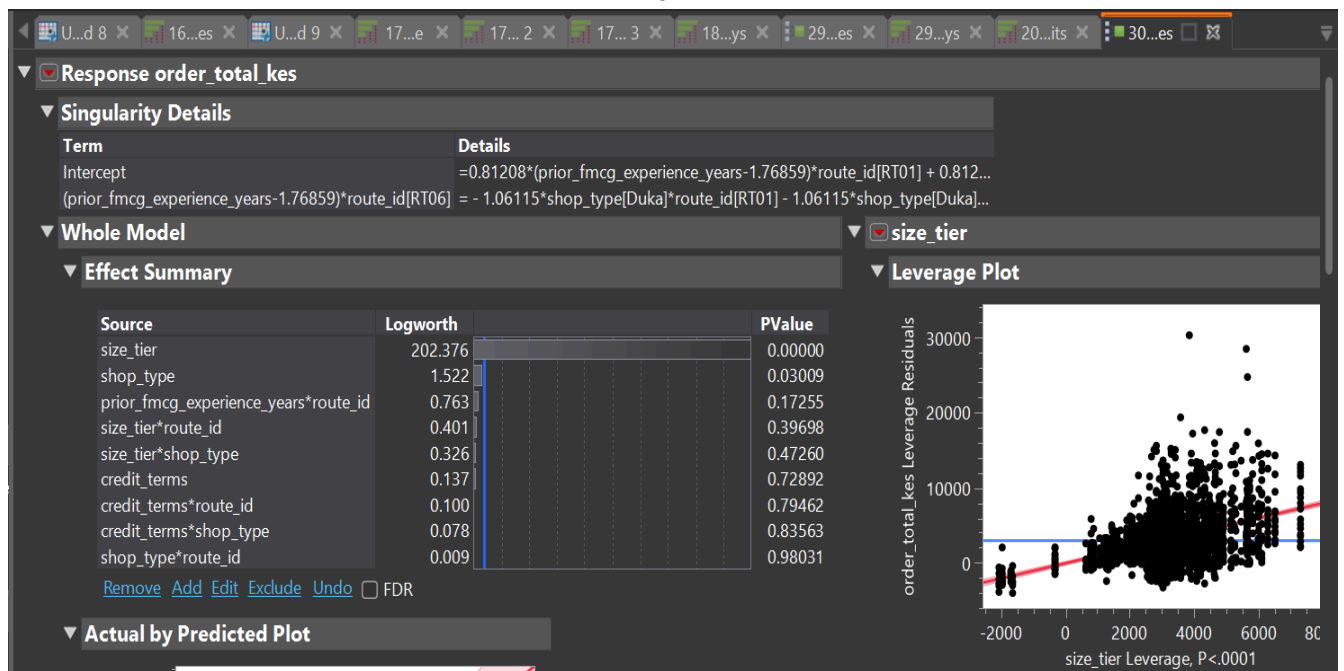
Predictor Screening [Copy Selected](#)

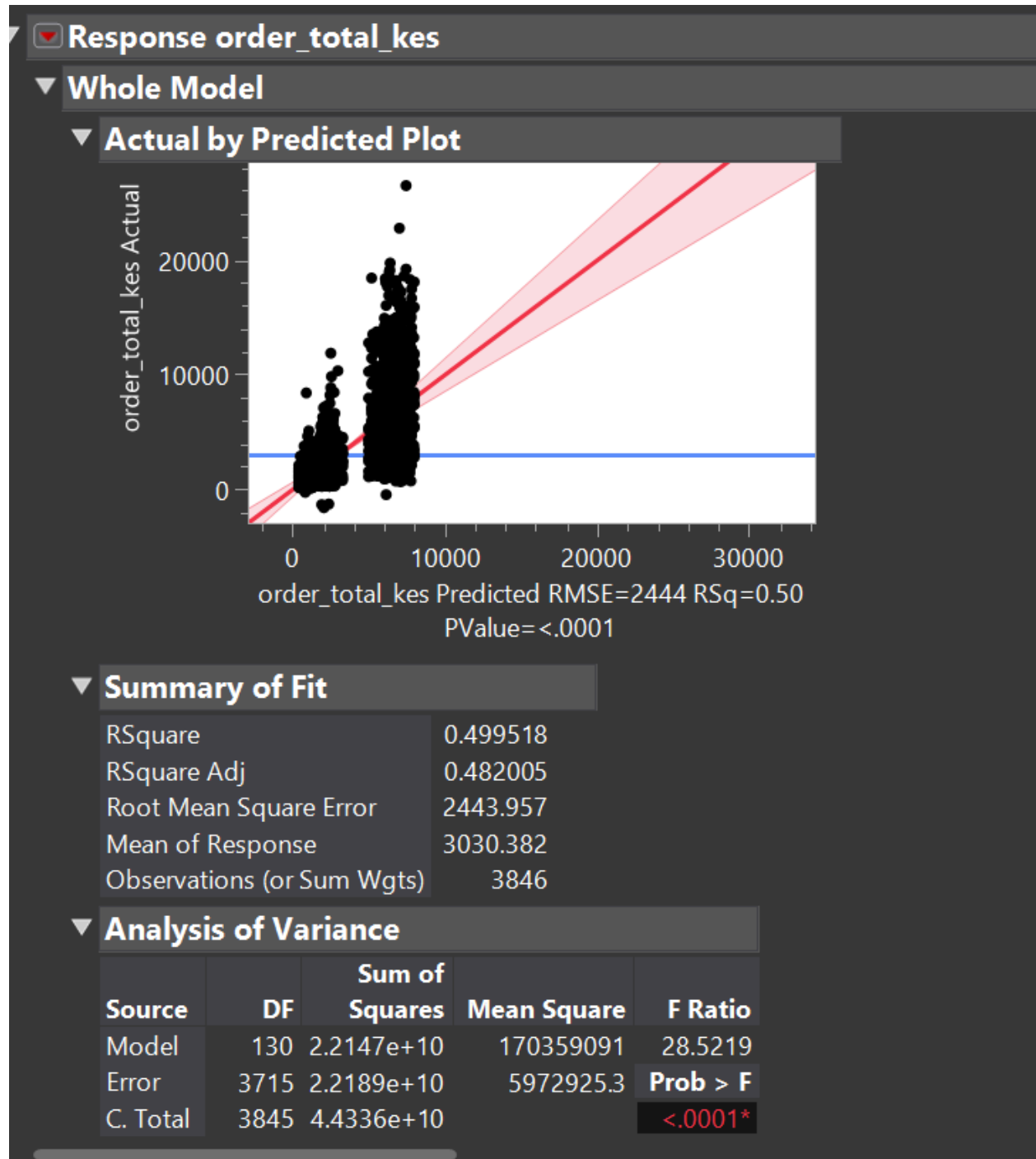
Predictor	Contribution	Portion	duration_days	Rank ^
category	28.2546	0.8116		1
cause	4.6859	0.1346		2
shelf_life_days	1.8711	0.0537		3
scope	0.0000	0.0000		4
cold_chain_required	0.0000	0.0000		5

Grain 6: Daily Inventory Spoilage (20_Inventory_Enriched_InventoryLevel)

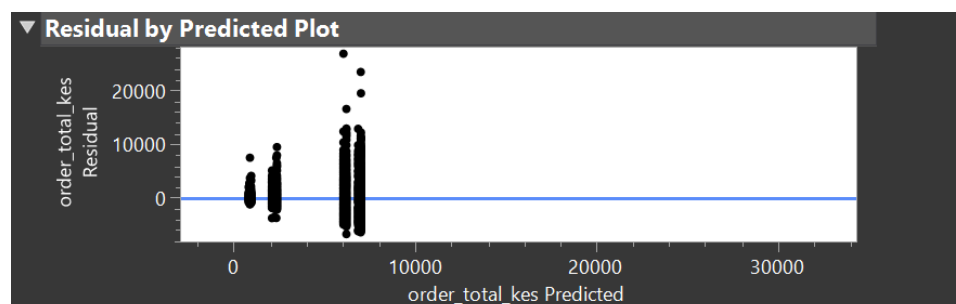
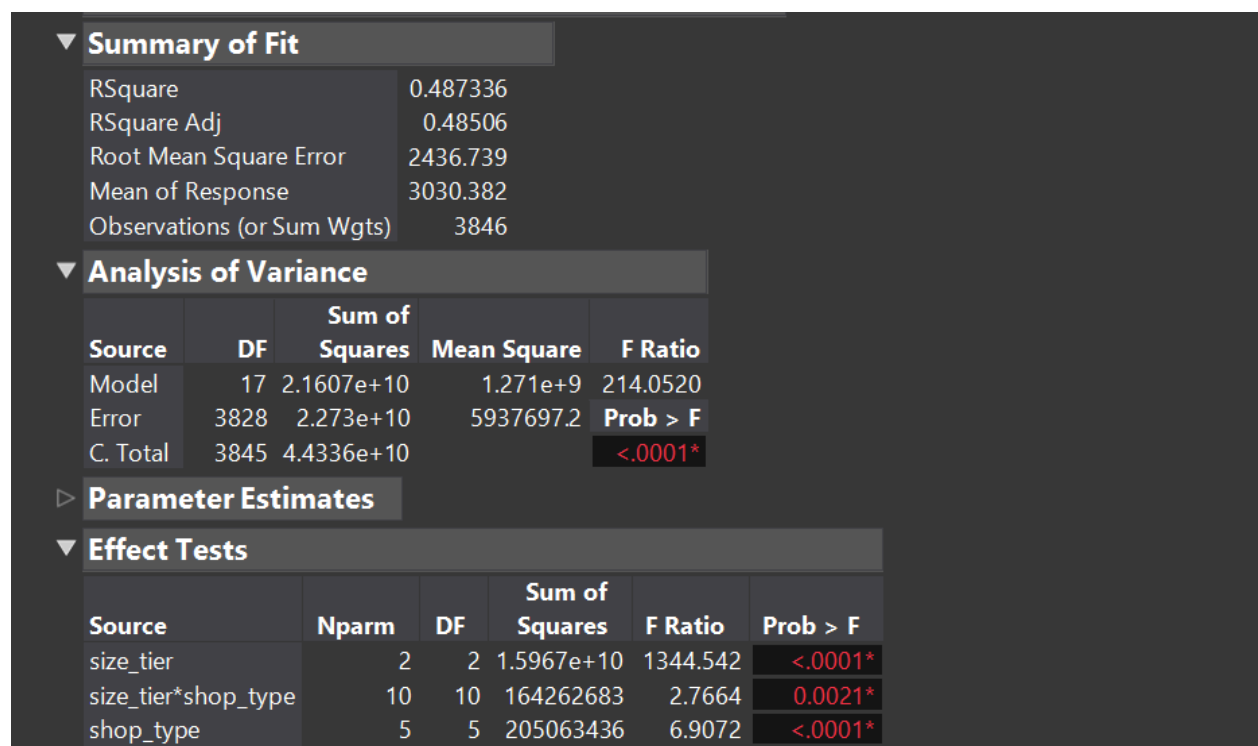
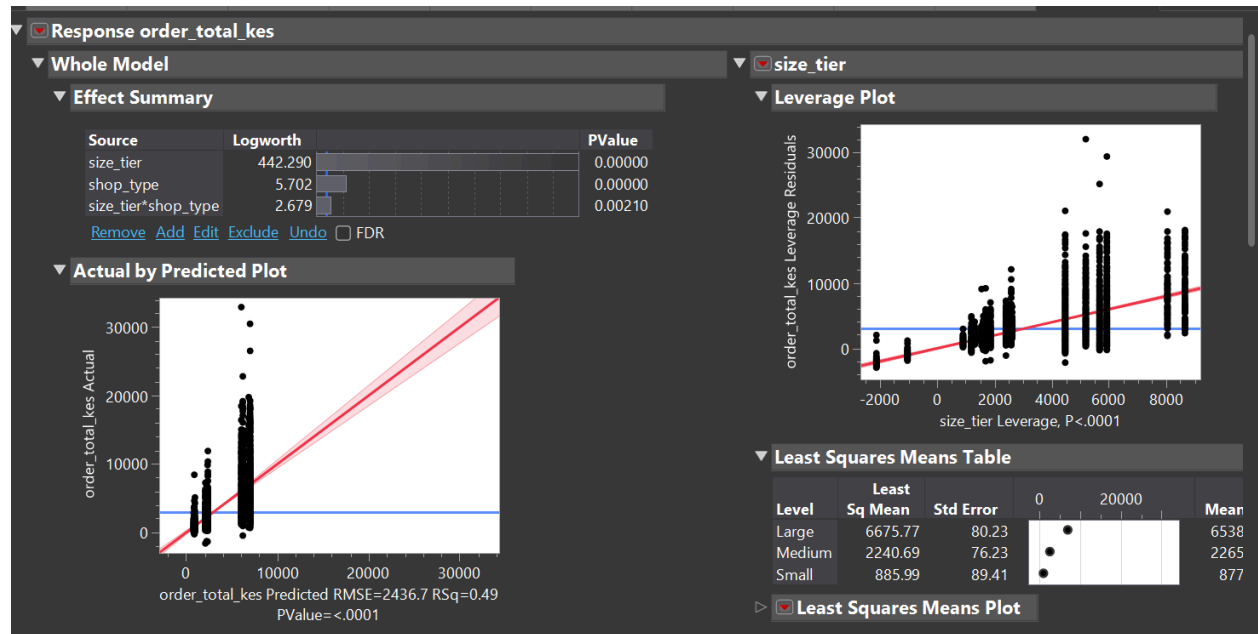
Predictor Screening				
Predictor	spoiled_units		Rank ^	
	Contribution	Portion		
opening_stock_units	13084.4	0.6161	1	
closing_stock_units	6420.1	0.3023	2	
sold_units	543.1	0.0256	3	
cold_room_temp_c	387.5	0.0182	4	
category	365.6	0.0172	5	
received_units	334.6	0.0158	6	
shelf_life_days	100.7	0.0047	7	
cold_chain_required	0.0	0.0000	8	

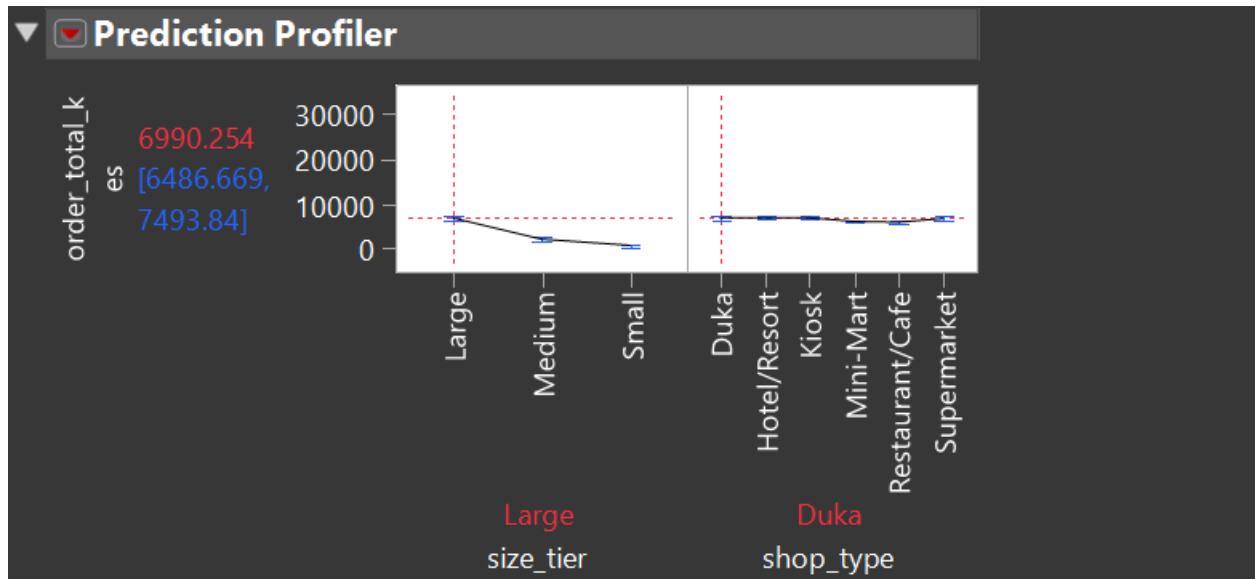
Model specification for Grain 1 (Order-Level Analysis)



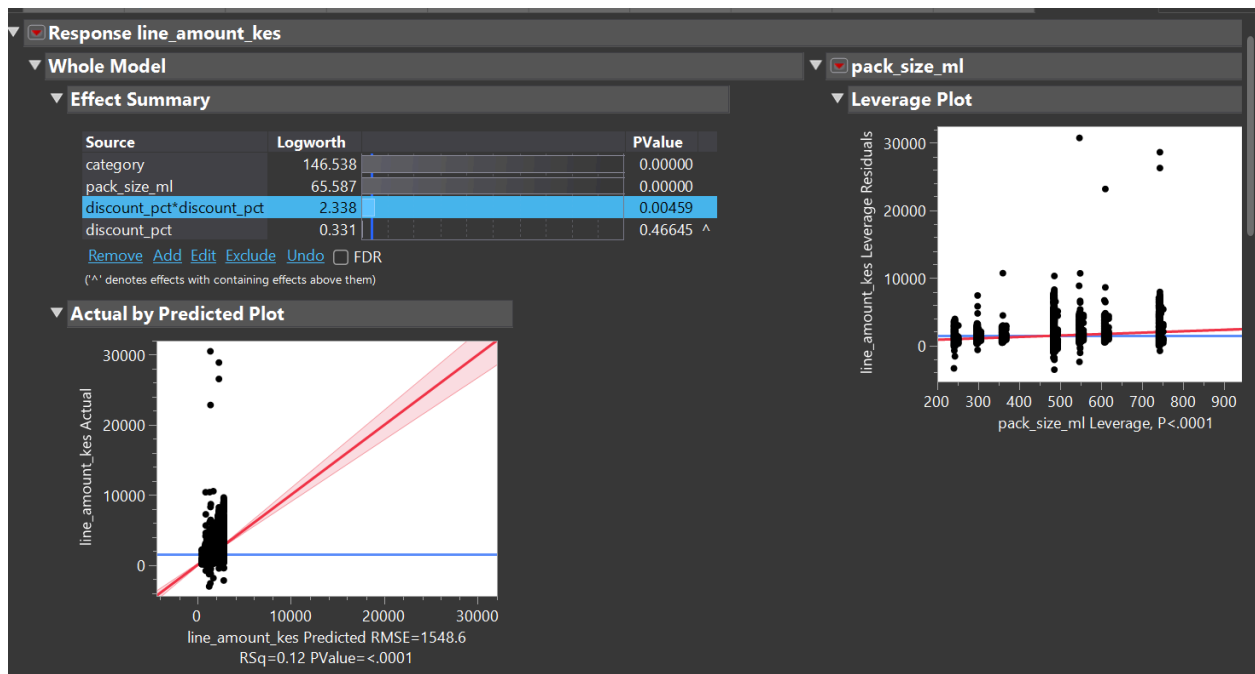


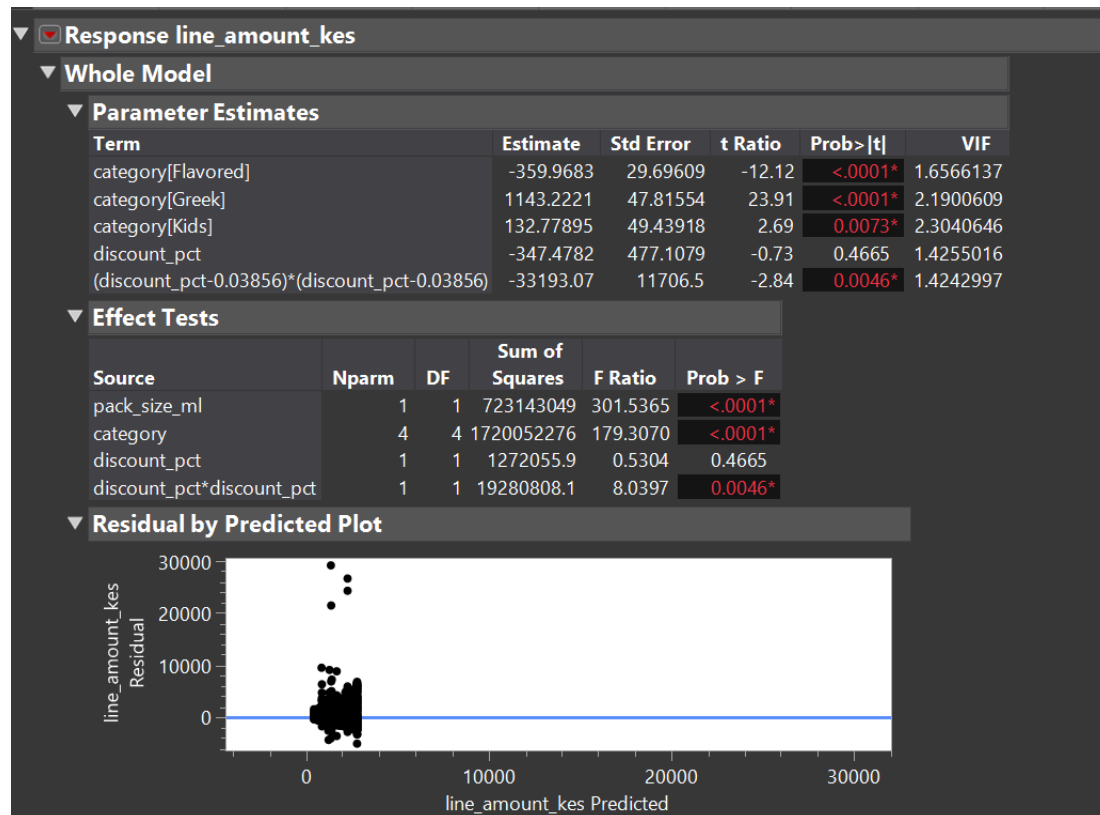
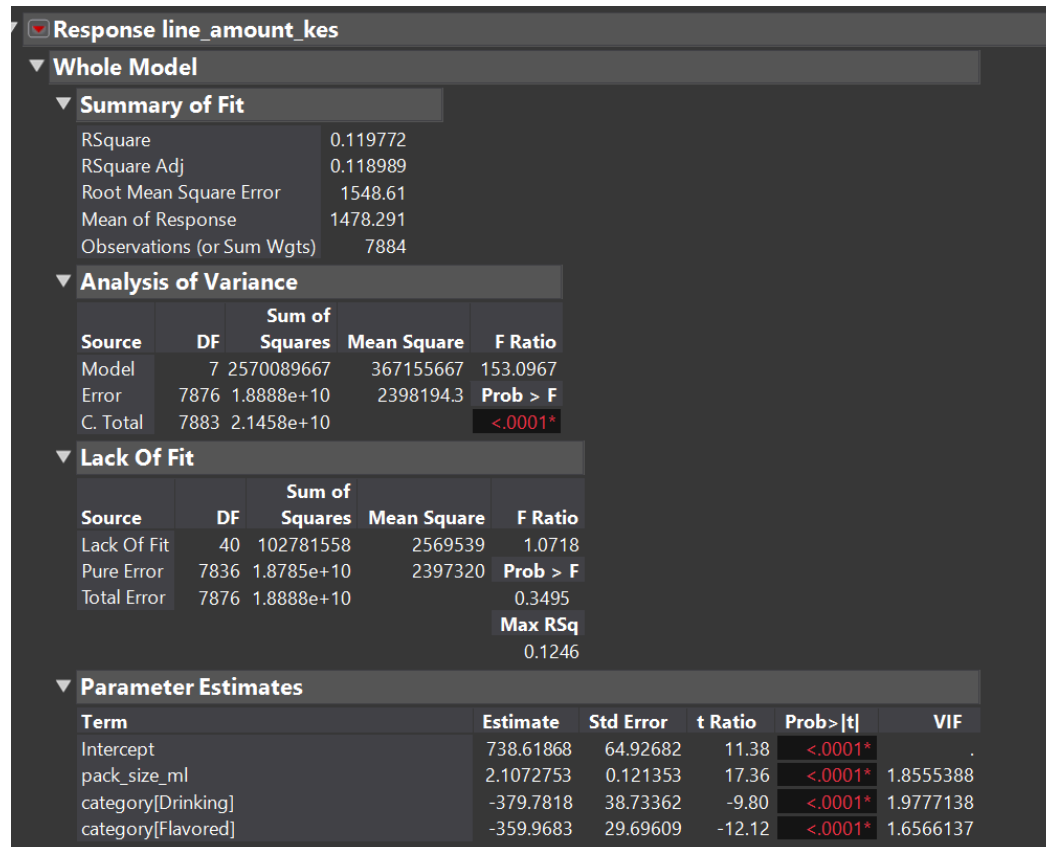
Reduced model for grain 1

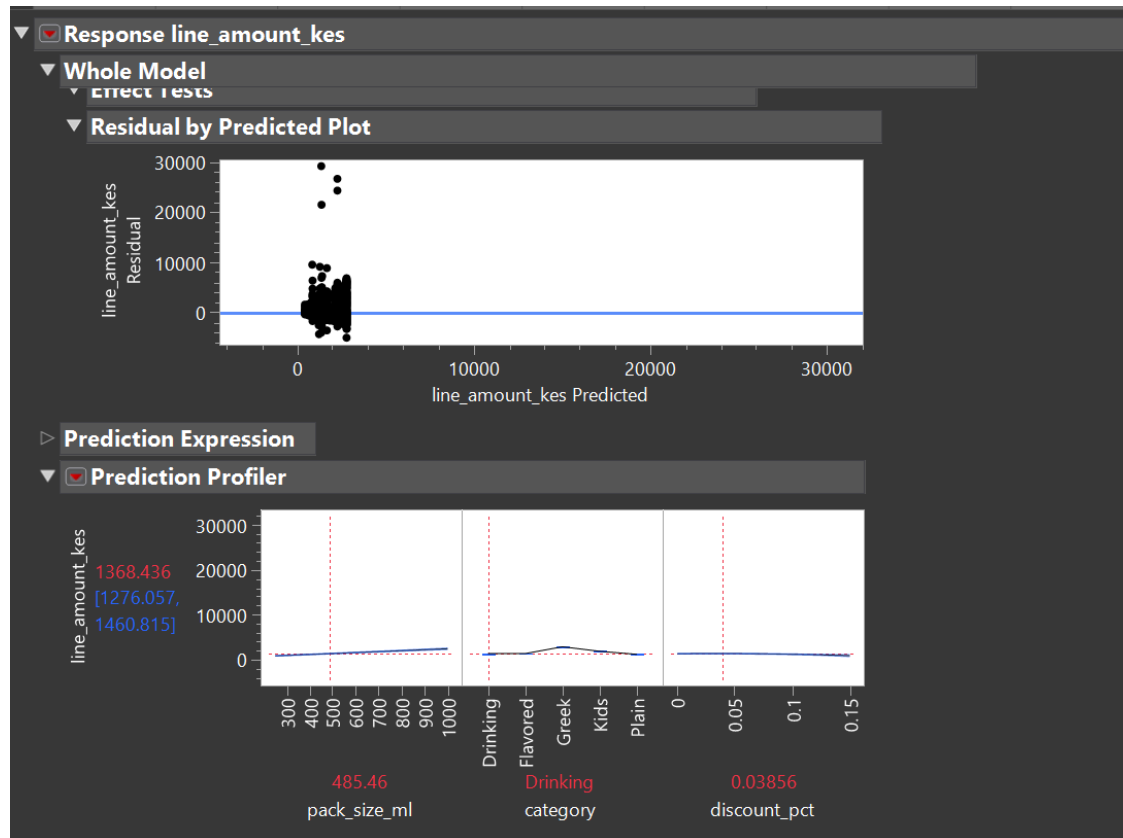




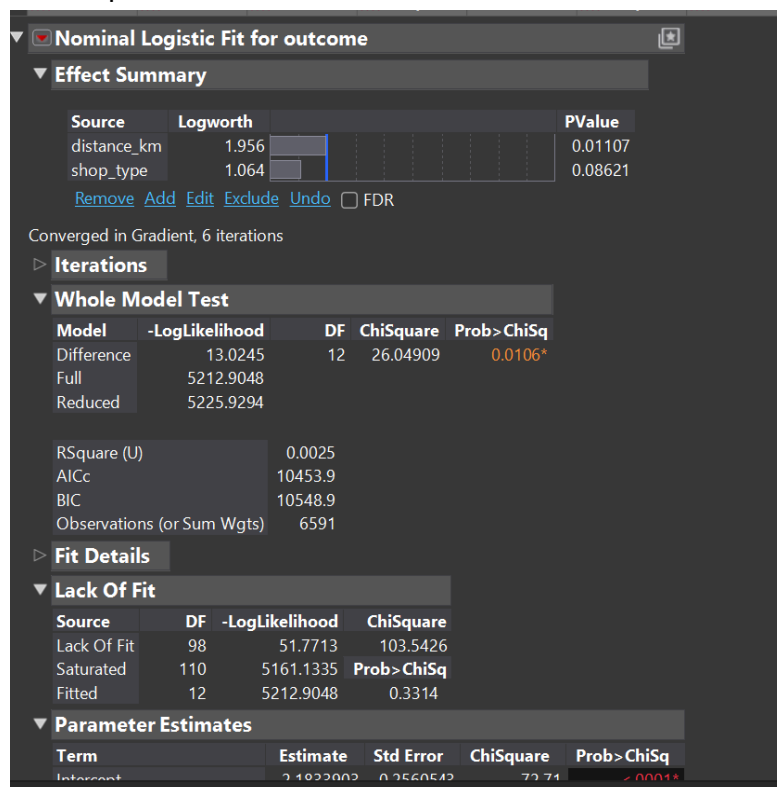
Model specification for Grain 2

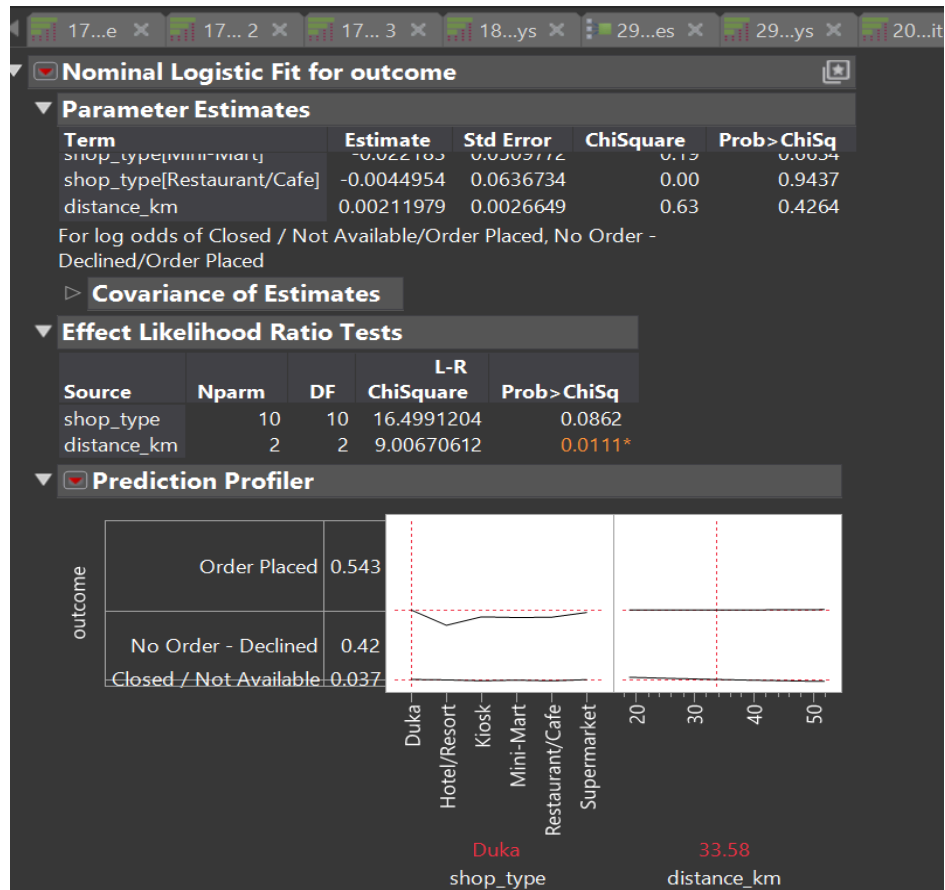
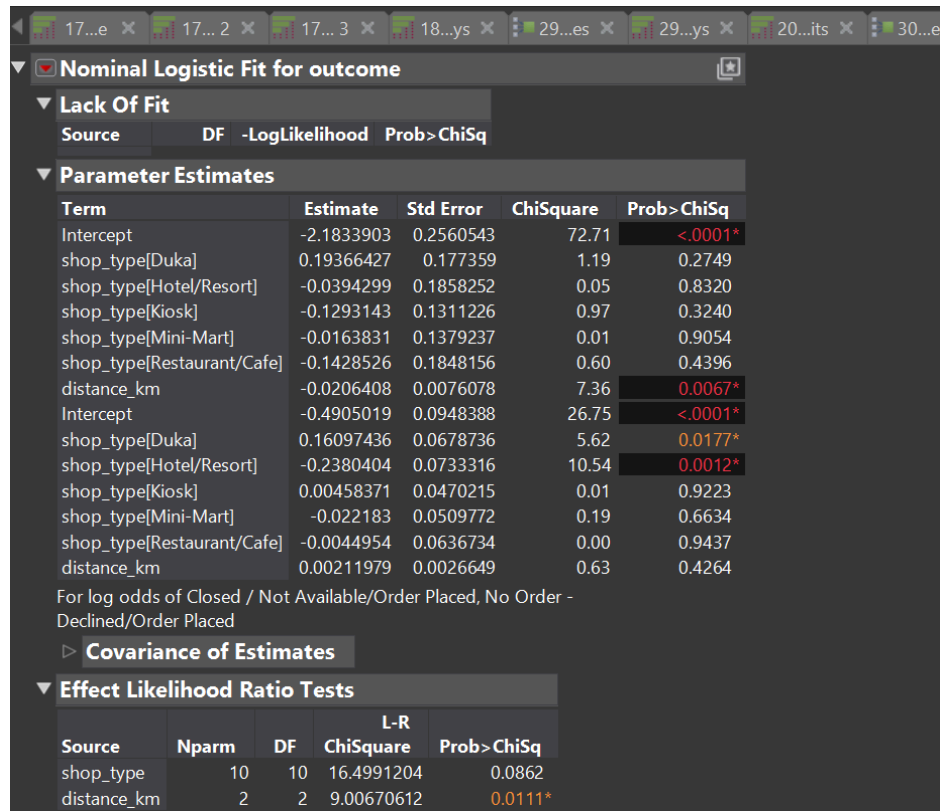




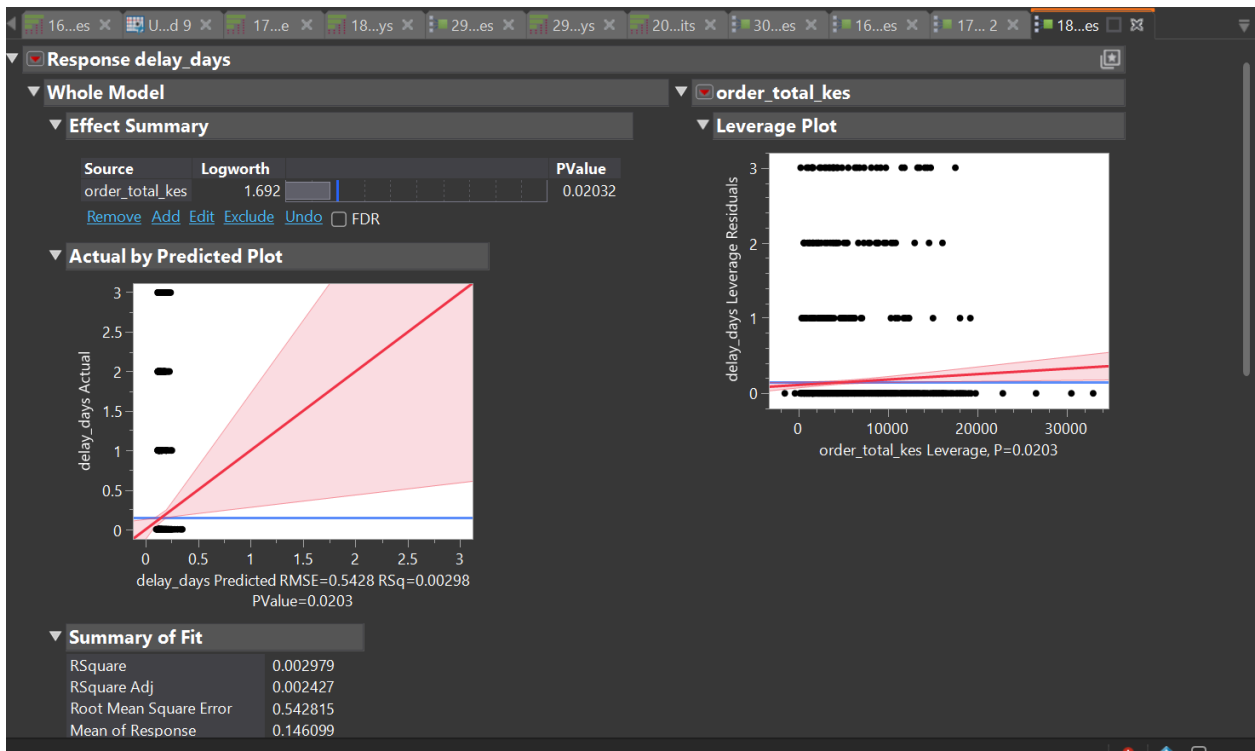


Model specification for Grain 3





Model specification for Grain 4-Reduced



Response delay_days

Whole Model

Summary of Fit

Statistic	Value
RSquare	0.002979
RSquare Adj	0.002427
Root Mean Square Error	0.542815
Mean of Response	0.146099
Observations (or Sum Wgts)	1807

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	1.58929	1.58929	5.3938
Error	1805	531.84070	0.29465	Prob > F
C. Total	1806	533.42999		0.0203*

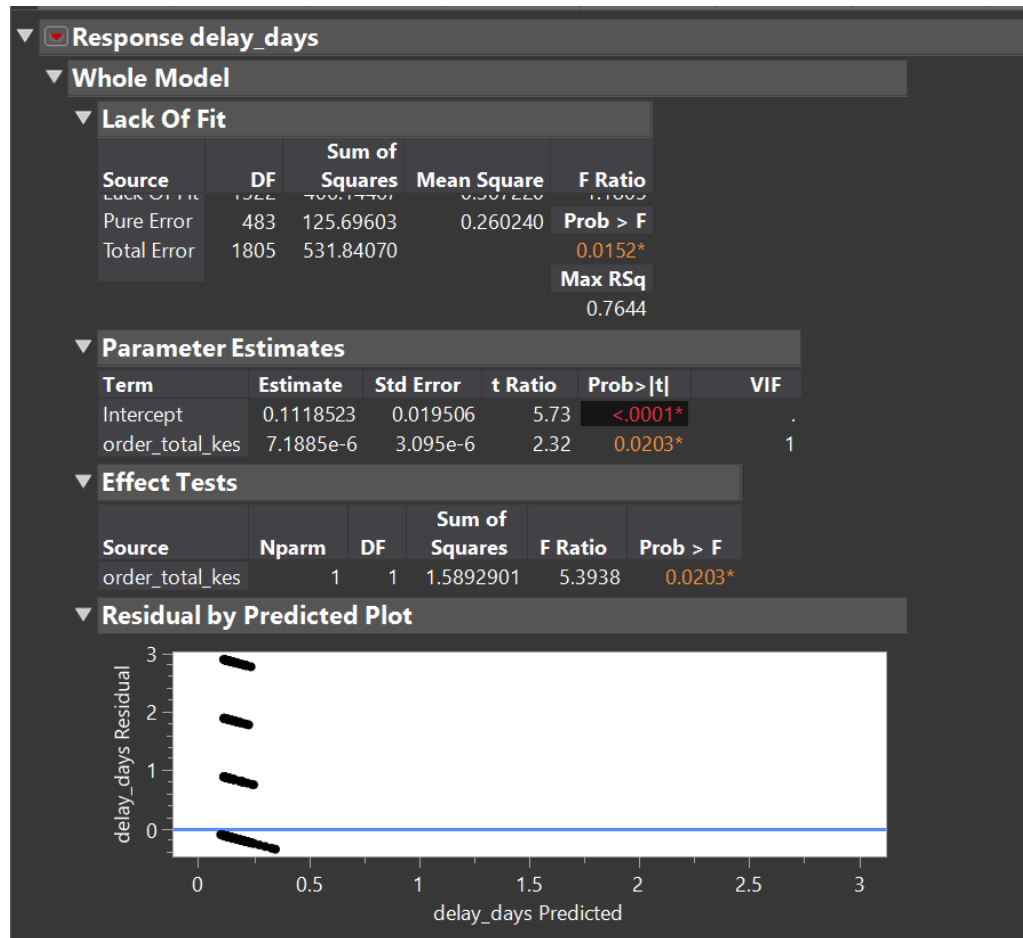
Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	1322	406.14467	0.307220	1.1805
Pure Error	483	125.69603	0.260240	Prob > F
Total Error	1805	531.84070		0.0152*

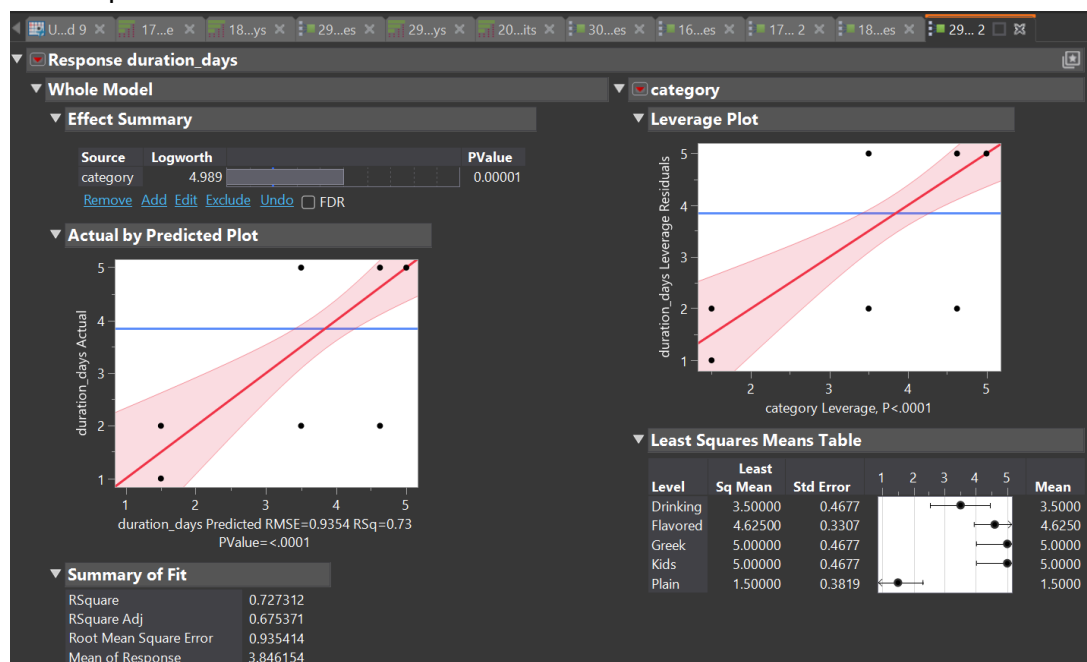
Max RSq
0.7644

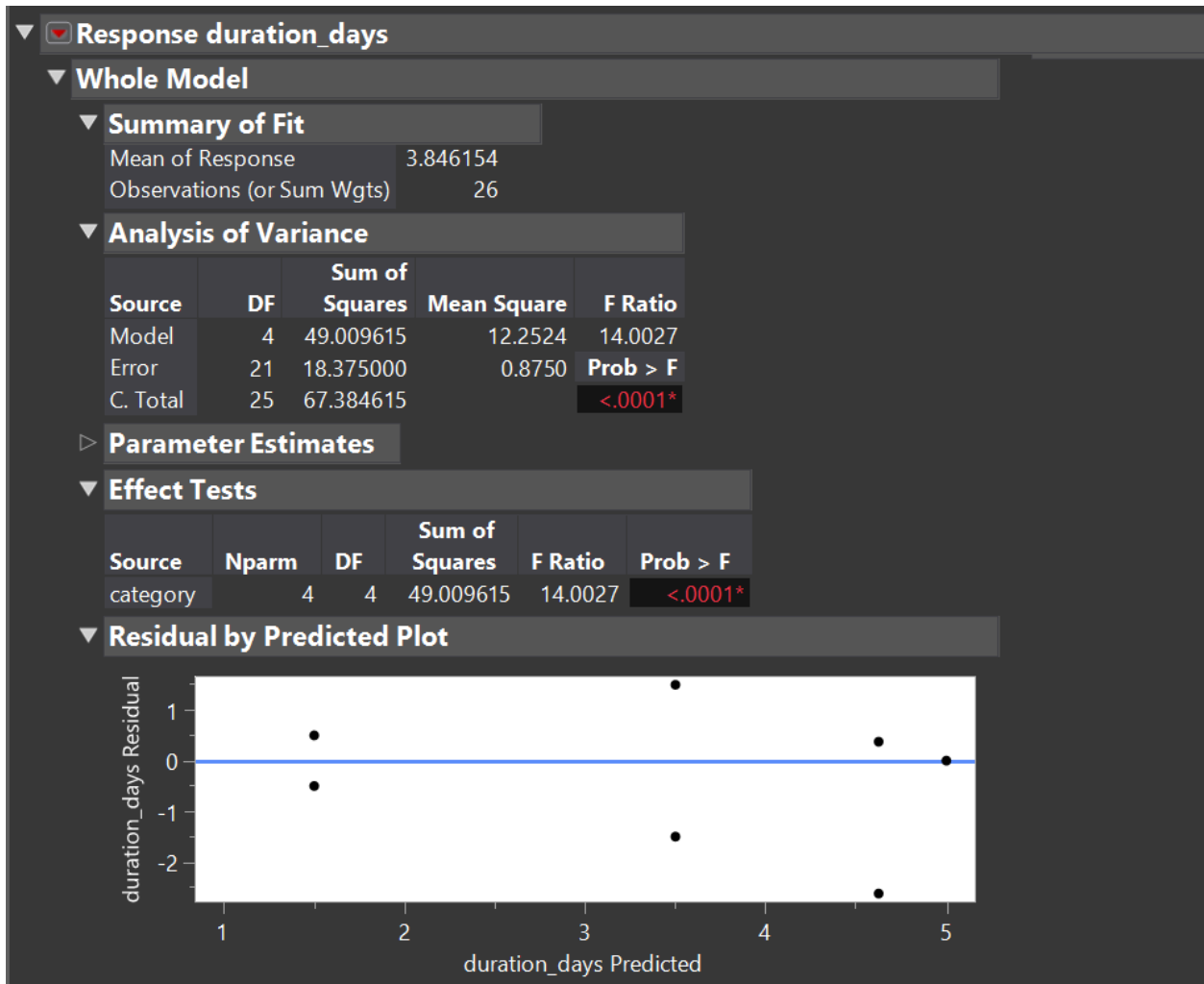
Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	0.1118523	0.019506	5.73	<.0001*	.
order_total_kes	7.1885e-6	3.095e-6	2.32	0.0203*	1

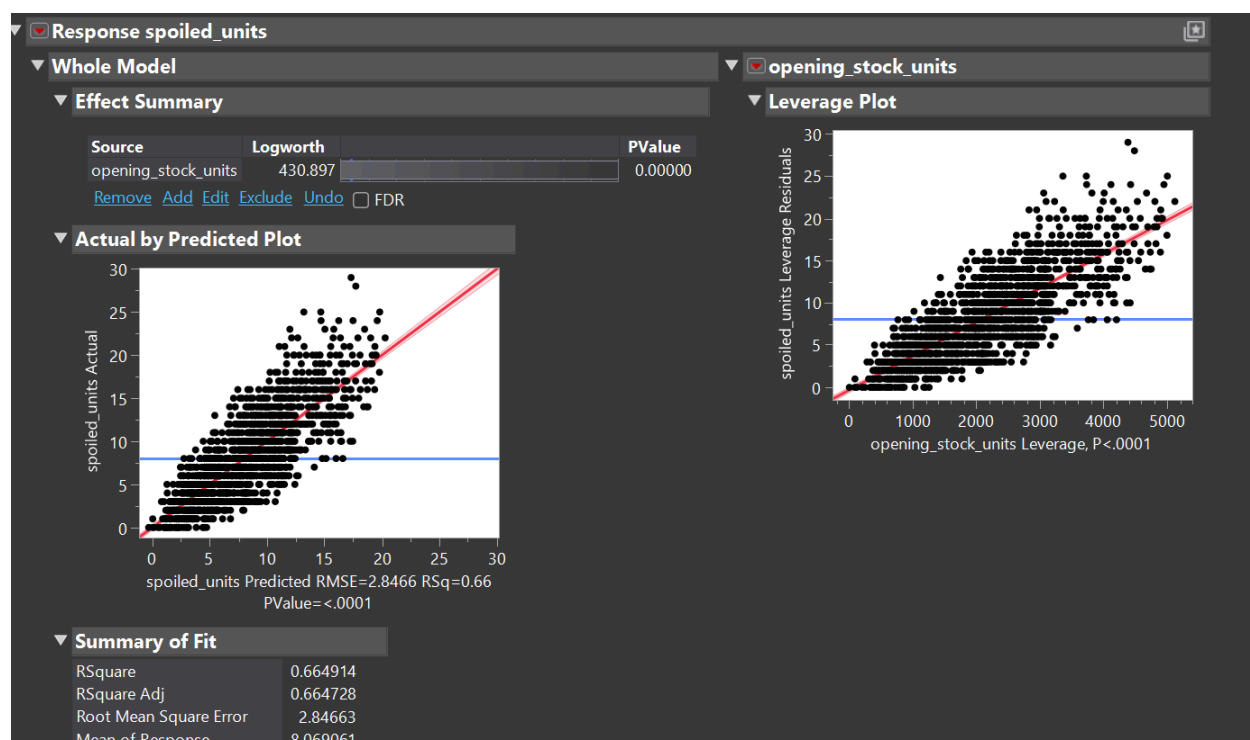


Model specification for Grain 5-Reduced





Model specification for **Grain 6**



Response spoiled_units

Whole Model

Summary of Fit

RSquare	0.664914
RSquare Adj	0.664728
Root Mean Square Error	2.84663
Mean of Response	8.069061
Observations (or Sum Wgts)	1810

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	29071.600	29071.6	3587.625
Error	1808	14650.767	8.1	Prob > F
C. Total	1809	43722.367		<.0001*

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-0.309479	0.155062	-2.00	0.0461*
opening_stock_units	0.0040205	6.712e-5	59.90	<.0001*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
opening_stock_units	1	1	29071.600	3587.625	<.0001*

Response spoiled_units

Whole Model

Summary of Fit

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	29071.600	29071.6	3587.625
Error	1808	14650.767	8.1	Prob > F
C. Total	1809	43722.367		<.0001*

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-0.309479	0.155062	-2.00	0.0461*
opening_stock_units	0.0040205	6.712e-5	59.90	<.0001*

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
opening_stock_units	1	1	29071.600	3587.625	<.0001*

Residual by Predicted Plot

